Practical Work

IT-M2-S1-Cyber: Network Intrusion Detection

TP 2: Creating Suricata Rules

Asma Saif

Objectives

- Identify suspicious elements by analyzing a PCAP.
- Write Suricata rules.
- Verify that a rule works correctly (non-regression testing).

Lab Environment

Suricata & Wireshark Installation

```
┌──(kali㉿kali)-[~]
└─$ suricata --version
suricata: unrecognized option '--version'
Suricata 8.0.3 ("undefined")
USAGE: suricata [OPTIONS] [BPF FILTER]
```

```
┌──(kali㉿kali)-[~]
└─$ wireshark --version
Wireshark 4.6.3.

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Licensed under the terms of the GNU General Public License (version 2 or late
r).
This is free software; see the file named COPYING in the distribution. There
is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE
.

Compile-time info:
 Bit width: 64-bit
  Compiler: GCC 15.2.0
```

Extracting PCAP

```
┌──(kali㉿kali)-[~/pcap]
└─$ wget https://www.malware-traffic-analysis.net/2024/12/04/2024-12-04-Agent
Tesla-variant-using-FTP.pcap.zip
--2026-03-06 04:07:22--  https://www.malware-traffic-analysis.net/2024/12/04/
2024-12-04-AgentTesla-variant-using-FTP.pcap.zip
Resolving www.malware-traffic-analysis.net (www.malware-traffic-analysis.net)
... 199.201.110.204
Connecting to www.malware-traffic-analysis.net (www.malware-traffic-analysis.
net)|199.201.110.204|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 18539 (18K) [application/zip]
Saving to: '2024-12-04-AgentTesla-variant-using-FTP.pcap.zip'

2024-12-04-AgentTes 100%[===================>]  18.10K  --.-KB/s    in 0.1s

2026-03-06 04:07:23 (138 KB/s) - '2024-12-04-AgentTesla-variant-using-FTP.pca
p.zip' saved [18539/18539]


┌──(kali㉿kali)-[~/pcap]
└─$ unzip 2024-12-04-AgentTesla-variant-using-FTP.pcap.zip
Archive:  2024-12-04-AgentTesla-variant-using-FTP.pcap.zip
[2024-12-04-AgentTesla-variant-using-FTP.pcap.zip] 2024-12-04-AgentTesla-vari
ant-using-FTP.pcap password:
  inflating: 2024-12-04-AgentTesla-variant-using-FTP.pcap

┌──(kali㉿kali)-[~/pcap]
└─$
```

Wireshark

4.1: Wireshark Traffic Analysis

Observed Protocol Hierarchy

Wireshark · Protocol Hierarchy Statistics · 2024-12-04-AgentTesla-variant-using-FTP.pcap

:ol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes
me	100.0	182	100.0	53464	352	0	0
Ethernet	100.0	182	4.8	2548	16	0	0
Internet Protocol Version 4	100.0	182	6.8	3640	24	0	0
User Datagram Protocol	3.3	6	0.1	48	0	0	0
Domain Name System	3.3	6	0.6	322	2	6	322
Transmission Control Protocol	96.7	176	87.7	46906	309	100	2112
Transport Layer Security	4.4	8	9.9	5287	34	7	2765
FTP Data	14.8	27	69.7	37289	246	24	0
Line-based text data	1.6	3	6.1	3286	21	3	3286
File Transfer Protocol (FTP)	23.1	42	3.9	2086	13	42	0

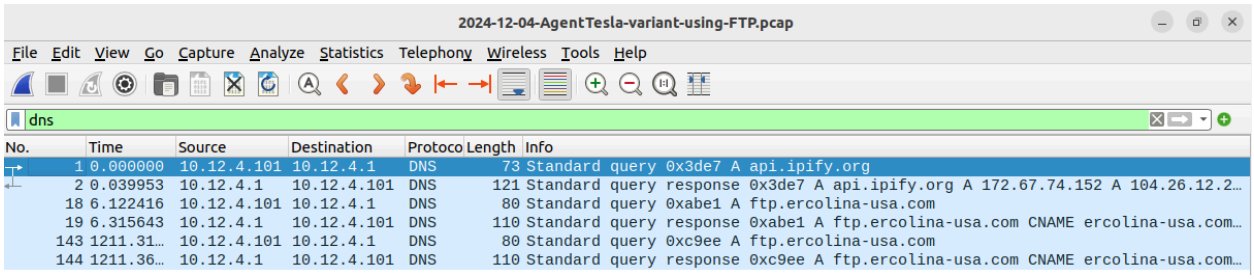
The protocol hierarchy shows that most traffic is carried over TCP, with some DNS activity over UDP. At the application layer, DNS, TLS, and FTP protocols are observed.
The presence of both FTP and FTP-DATA traffic is particularly significant, as it indicates file transfers between the infected host and a remote server. The high proportion of FTP-DATA traffic suggests that actual file content is being transmitted, which is consistent with data exfiltration behavior.

The analysis of the PCAP file using Wireshark reveals network activity generated by a compromised host communicating with external servers.

The internal IP address 10.12.4.101 appears to be the infected machine. It initiates multiple connections to external IP addresses over different protocols, including DNS, TLS, and FTP.
The traffic shows a clear sequence of actions:

1. Domain resolution via DNS
2. Encrypted communication via TLS
3. File exfiltration via FTP

Filtering Packets with DNS:



No.	Time	Source	Destination	Protoco	Length	Info
1	0.000000	10.12.4.101	10.12.4.1	DNS	73	Standard query 0x3de7 A api.ipify.org
2	0.039953	10.12.4.1	10.12.4.101	DNS	121	Standard query response 0x3de7 A api.ipify.org A 172.67.74.152 A 104.26.12.2…
18	6.122416	10.12.4.101	10.12.4.1	DNS	80	Standard query 0xabe1 A ftp.ercolina-usa.com
19	6.315643	10.12.4.1	10.12.4.101	DNS	110	Standard query response 0xabe1 A ftp.ercolina-usa.com CNAME ercolina-usa.com…
143	1211.31…	10.12.4.101	10.12.4.1	DNS	80	Standard query 0xc9ee A ftp.ercolina-usa.com
144	1211.36…	10.12.4.1	10.12.4.101	DNS	110	Standard query response 0xc9ee A ftp.ercolina-usa.com CNAME ercolina-usa.com…

We can clearly see two domains being resolved:

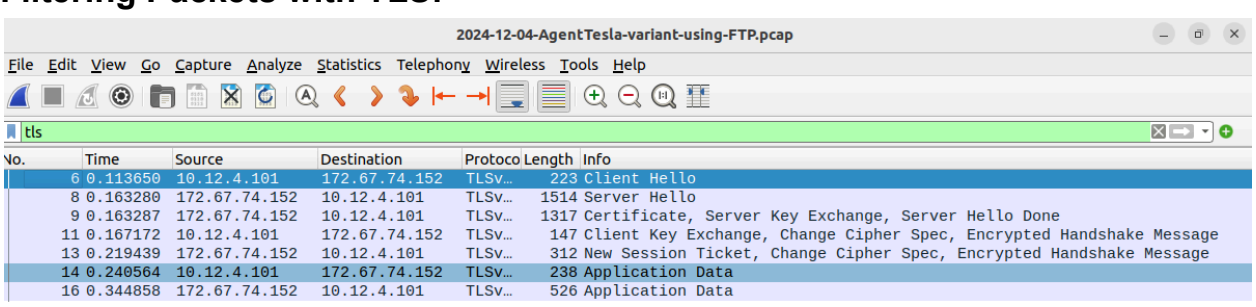
- api.ipify.org — the malware checking its public IP address
- ftp.ercolina-usa.com — the malicious FTP server where stolen data is uploaded

Key IPs visible:

- 10.12.4.101 — the infected machine
- 10.12.4.1 — the DNS server

The DNS traffic shows that the infected host (10.12.4.101) performs domain name resolution before establishing connections. The request to api.ipify.org is commonly used by malware to determine the public IP address of the infected machine.
The domain ftp.ercolina-usa.com resolves to an external server used later for FTP communication, indicating a potential command-and-control or data exfiltration server

Filtering Packets with TLS:



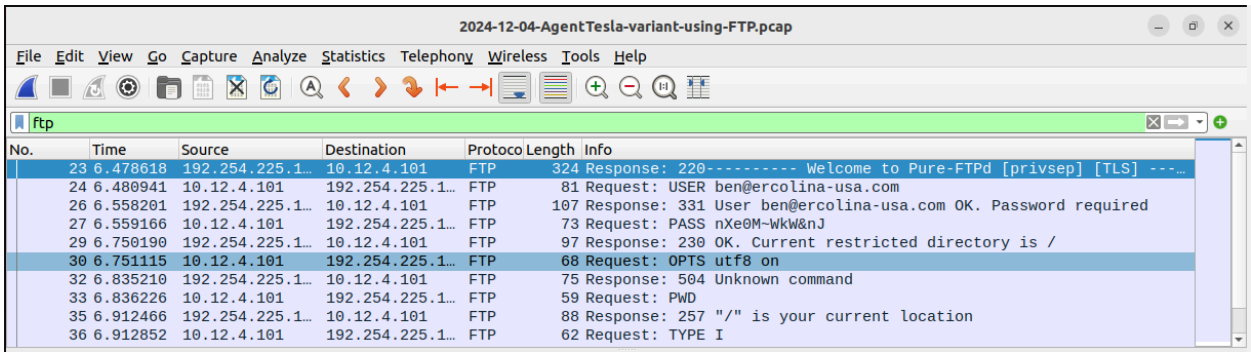
No.	Time	Source	Destination	Protoco	Length	Info
6	0.113650	10.12.4.101	172.67.74.152	TLSv…	223	Client Hello
8	0.163280	172.67.74.152	10.12.4.101	TLSv…	1514	Server Hello
9	0.163287	172.67.74.152	10.12.4.101	TLSv…	1317	Certificate, Server Key Exchange, Server Hello Done
11	0.167172	10.12.4.101	172.67.74.152	TLSv…	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
13	0.219439	172.67.74.152	10.12.4.101	TLSv…	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
14	0.240564	10.12.4.101	172.67.74.152	TLSv…	238	Application Data
16	0.344858	172.67.74.152	10.12.4.101	TLSv…	526	Application Data

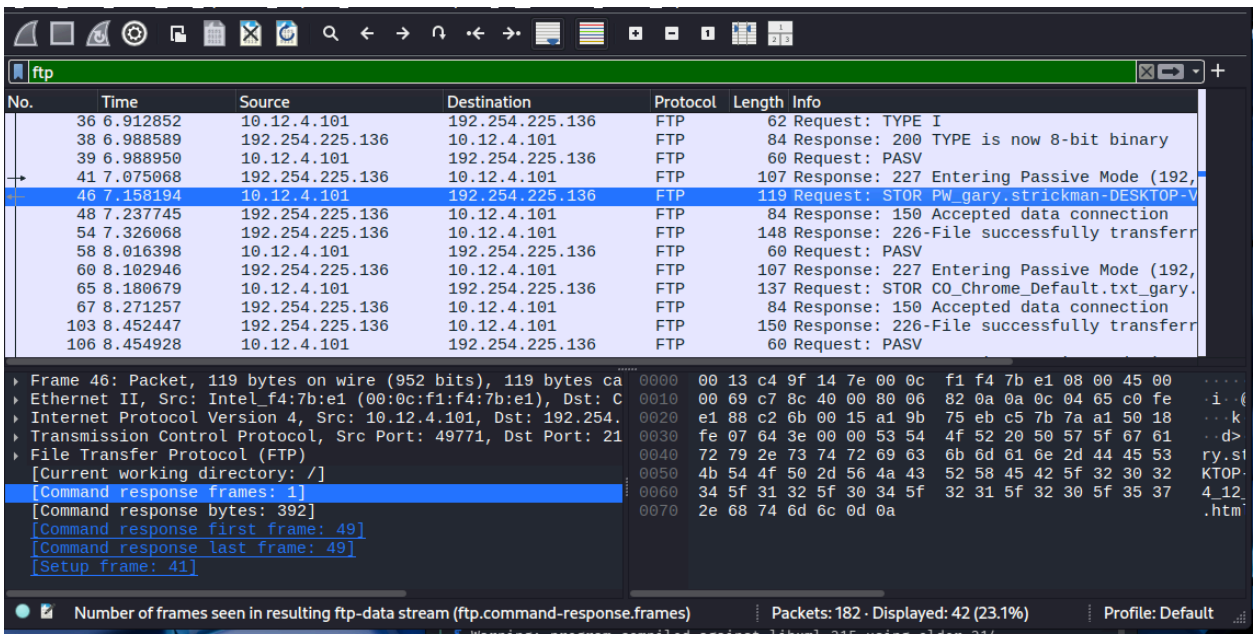
The TLS traffic shows encrypted communication between the infected host (10.12.4.101) and the external IP address 172.67.74.152, which corresponds to the domain api.ipify.org.
The TLS handshake includes:

- Client Hello
- Server Hello
- Certificate exchange
- Encrypted application data

This confirms that the infected machine establishes a secure connection to retrieve its public IP address.

Filtering Packets with FTP:





You can see clearly:

- Packet 46: STOR PW_gary.strickman-DESKTOP-V...uploading the HTML credentials file
- Packet 48: Response: 150 Accepted data connection
- Packet 54: Response: 226 File successfully transferred
- Packet 65: STOR CO_Chrome_Default.txt_gary... — another file being uploaded

Also in the bottom panel the full filename is partially visible in the hex dump.

This traffic reveals clear evidence of data exfiltration from the infected host. The infected machine (10.12.4.101) connects to the external FTP server 192.254.225.136 (resolved from ftp.ercolina-usa.com).

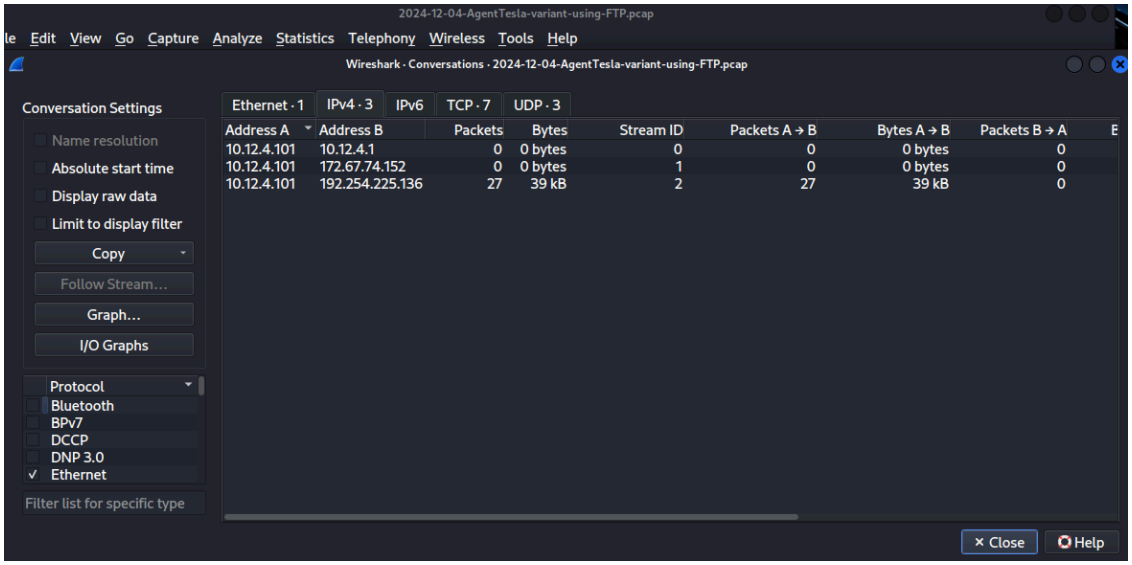
Observed behavior:

- Successful authentication:
  - Username: ben@ercolina-usa.com
  - Password: nXeOM-WKW&nJ
- Multiple file uploads using the STOR command

The FTP server responds with:

- 150 Accepted data connection
- 226 File successfully transferred

This confirms that the files were successfully exfiltrated.



- **10.12.4.101** ↔ **10.12.4.1**: DNS queries to local DNS server
- **10.12.4.101** ↔ **172.67.74.152**: TLS connection to api.ipify.org (public IP check)
- **10.12.4.101** ↔ **192.254.225.136**: FTP exfiltration to ftp.ercolina-usa.com (39 kB of stolen data sent)

In summary, the PCAP contains traffic from an infected machine 10.12.4.101. The malware first performs a DNS query to resolve api.ipify.org (resolved to 172.67.74.152) — this is a public IP lookup service used by Agent Tesla to identify the victim's public IP address. A TLS connection is then established to that IP. The malware also resolves ftp.ercolina-usa.com (resolved to 192.254.225.136) which is the attacker's FTP server used for data exfiltration. The infected machine then connects via FTP to this server and uploads stolen credential files using the STOR command, with the actual file contents transferred over FTP-DATA (approximately 39 kB of stolen data).

Wireshark · Follow TCP Stream (tcp.stream eq 1) · 2024-12-04-AgentTesla-variant-using-FTP.pcap

```
220---------- Welcome to Pure-FTPd [privsep] [TLS] ----------
220-You are user number 2 of 150 allowed.
220-Local time is now 16:21. Server port: 21.
220-IPv6 connections are also welcome on this server.
220 You will be disconnected after 15 minutes of inactivity.
USER ben@ercolina-usa.com
331 User ben@ercolina-usa.com OK. Password required
PASS nXe0M~WkW&nJ
230 OK. Current restricted directory is /
OPTS utf8 on
504 Unknown command
PWD
257 "/" is your current location
TYPE I
200 TYPE is now 8-bit binary
PASV
227 Entering Passive Mode (192,254,225,136,181,153)
STOR PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html
150 Accepted data connection
226-File successfully transferred
226 0.087 seconds (measured here), 4.39 Kbytes per second
PASV
227 Entering Passive Mode (192,254,225,136,147,201)
STOR CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt
150 Accepted data connection
226-File successfully transferred
226 0.184 seconds (measured here), 126.77 Kbytes per second
PASV
227 Entering Passive Mode (192,254,225,136,175,147)
STOR CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-
VJCRXEB_2024_12_04_21_21_04.txt
150 Accepted data connection
226-File successfully transferred
```

Using the “Follow TCP Stream” feature in Wireshark, we can reconstruct the full FTP session. The stream clearly shows:

- Credentials transmitted in clear text
- FTP commands and server responses
- File names being uploaded

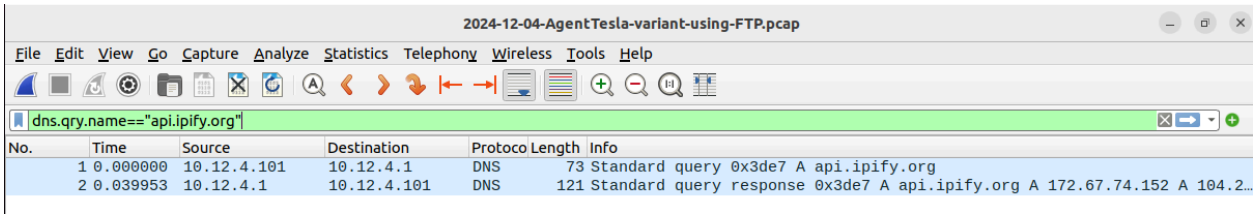
This confirms that:

- The malware does not use encryption for exfiltration
- Sensitive data is transmitted in plain text
- The FTP server is actively receiving stolen data

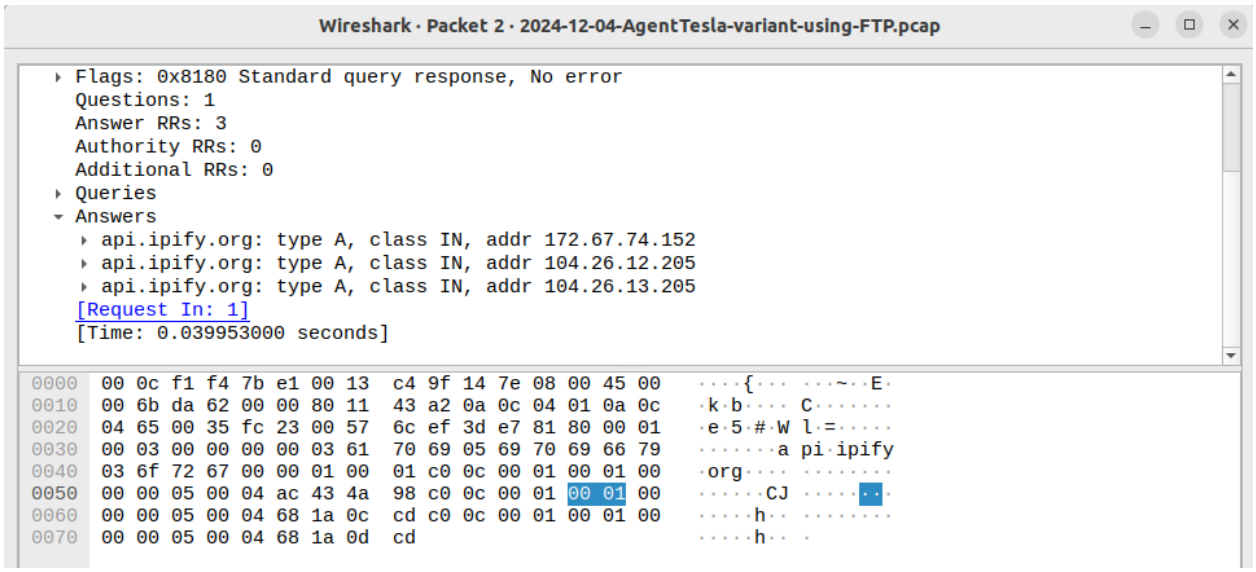
Suricata Rule Development

5.1: Write a rule that detects any TLS traffic destined for "api.ipify.org"

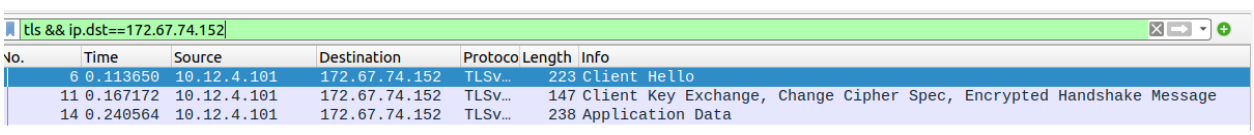
Using the filter to show only DNS packets where the queried domain name is exactly api.ipify.org

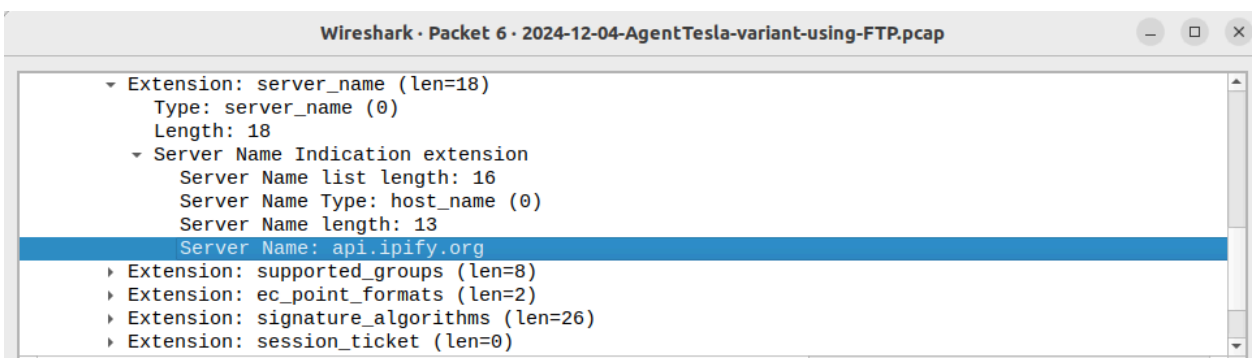


DNS Response packet:



The malware queries api.ipify.org to identify the public IP of the infected machine. Capturing the DNS resolution helps us identify the exact IP destination for the TLS connection.





TLS handshakes to api.ipify.org indicate the malware is trying to reach an external service to discover its public IP. Observing SNI in the Client Hello confirms the target domain.

Suricata Rule:

```
alert tls any any -> any any (msg:"Agent Tesla TLS connection to
api.ipify.org"; tls.sni; content:"api.ipify.org"; nocase;
classtype:network-scan; reference:url,api.ipify.org;
metadata:created_at 2026_03_19, confidence high, signature_severity
low, mitre_tactic_id TA0007, mitre_tactic_name Discovery,
mitre_technique_id T1016, mitre_technique_name System Network
Configuration Discovery, status testing; sid:1000001; rev:2;)
```

This rule detects TLS connections to api.ipify.org by inspecting the Server Name Indication (SNI) field in the TLS handshake. This domain is commonly used by malware to determine the infected host's public IP address. The rule is classified as network-scan since it represents reconnaissance activity. MITRE ATT&CK mapping corresponds to Discovery (TA0007), specifically System Network Configuration Discovery (T1016). The rule has high confidence due to the specificity of the domain.

Results:

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
12/04/2024-16:20:56.211571  [**] [1:1000002:1] Agent Tesla FTP file upload (STOR) [**] [Classification: (
null)] [Priority: 3] {TCP} 10.12.4.101:49771 -> 192.254.225.136:21
12/04/2024-16:20:56.375191  [**] [1:1000001:2] Agent Tesla TLS connection to api.ipify.org [**] [Classifi
cation: Detection of a Network Scan] [Priority: 3] {TCP} 10.12.4.101:49770 -> 172.67.74.152:443
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2024-12-04T16:20:56.375191-0500",
  "flow_id": 275566959779929,
  "pcap_cnt": 10,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49770,
  "dest_ip": "172.67.74.152",
  "dest_port": 443,
  "proto": "TCP",
```

```
"event_type": "alert",
"src_ip": "10.12.4.101",
"src_port": 49770,
"dest_ip": "172.67.74.152",
"dest_port": 443,
"proto": "TCP",
"ip_v": 4,
"pkt_src": "wire/pcap",
"tx_id": 0,
"alert": {
  "action": "allowed",
  "gid": 1,
  "signature_id": 1000001,
  "rev": 2,
  "signature": "Agent Tesla TLS connection to api.ipify.org",
  "category": "Detection of a Network Scan",
  "severity": 3,
  "metadata": {
    "confidence": [
      "high"
    ],
    "created_at": [
      "2026_03_19"
    ],
    "mitre_tactic_id": [
      "TA0007"
    ],
    "mitre_tactic_name": [
      "Discovery"
    ],
    "mitre_technique_id": [
      "T1016"
    ],
```

```
"tls": {
  "subject": "CN=ipify.org",
  "issuerdn": "C=US, O=Google Trust Services, CN=WE1",
  "subjectaltname": [
    "ipify.org",
    "*.ipify.org"
  ],
  "serial": "00:D3:65:EE:22:80:2B:FF:E9:0E:7A:F5:A9:7A:CC:92:B1",
  "fingerprint": "d4:6a:1f:31:16:85:d7:9c:b8:93:15:4f:53:86:d8:a7:4d:15:f8:64",
  "sni": "api.ipify.org",
  "version": "TLS 1.2",
  "notbefore": "2024-11-13T05:59:28",
  "notafter": "2025-02-11T05:59:27",
  "ja4": "t12d180700_4b22cbed5bed_2dae41c691ec"
},
```

```
"app_proto": "ftp",
"direction": "to_server",
"flow": {
  "pkts_toserver": 21,
  "pkts_toclient": 27,
  "bytes_toserver": 1475,
  "bytes_toclient": 2447,
  "start": "2024-12-04T16:21:02.528772-0500",
  "src_ip": "10.12.4.101",
  "dest_ip": "192.254.225.136",
  "src_port": 49771,
  "dest_port": 21
}
```


The Suricata execution generated two alerts, confirming that both custom rules were successfully triggered. From the fast.log, the first alert shows that the internal host 10.12.4.101 initiated an FTP connection to 192.254.225.136 on port 21. The rule detecting the STOR command was triggered, indicating a file upload operation. This behavior is consistent with data exfiltration, where the malware uploads stolen information to a remote server.

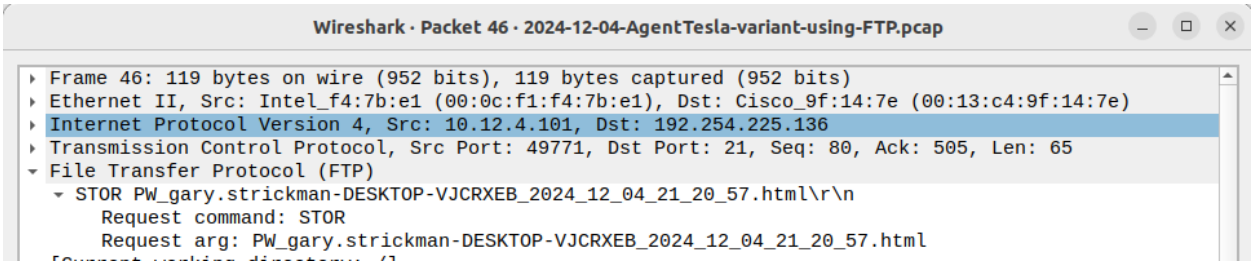
The second alert corresponds to a TLS connection from the same internal host to 172.67.74.152 on port 443. The rule matched the domain api.ipify.org, confirming that the host attempted to retrieve its public IP address. This is a known behavior of malware performing environment discovery before exfiltration.

The detailed eve.json output provides additional context. It confirms that the TLS handshake was successfully completed and reveals the Server Name Indication (SNI) value api.ipify.org, which triggered the rule. It also includes certificate information and flow statistics, such as the number of packets and bytes exchanged.

Flow data shows limited traffic for the TLS session, consistent with a simple request-response exchange, while the FTP session involves more packets, indicating an active file transfer.

Finally, the presence of metadata (e.g., MITRE tactic Discovery (TA0007) and technique System Network Configuration Discovery (T1016)) enriches the alert, helping classify the observed behavior within a known attack framework.

5.2: Write a rule to detect the upload of the file "PW_gary.strickman.." over the FTP protocol



The FTP STOR (upload file to server) command is used by a client to upload a file to a remote server. In this traffic, the infected host uploads a file containing stolen data. This confirms that the malware is actively exfiltrating information.

Suricata Rule:

```
alert ftp any any -> any any (msg:"Agent Tesla FTP Upload HTML Exfil File"; flow:established,to_server; content:"STOR"; nocase; content:".html"; distance:0; classtype:trojan-activity; reference:url,malpedia.caad.fkie.fraunhofer.de/details/win.agent_tesla; sid:1000002; rev:1; metadata:created_at 2024_12_04, confidence high, signature_severity major, mitre_tactic_id TA0010, mitre_tactic_name Exfiltration, mitre_technique_id T1048, mitre_technique_name Exfiltration_Over_Alternative_Protocol, status testing;)
```

This Suricata rule is designed to detect the exfiltration of HTML files over FTP by the Agent Tesla malware. It monitors all FTP traffic in both directions (any any -> any any) and targets established connections flowing toward the server (flow:established,to_server), meaning it focuses on upload activity initiated by the infected machine. The rule looks for the FTP STOR command (case-insensitive via nocase), which is the command a client sends to instruct the server to store i.e., receive a file. Immediately following that, it checks for the .html extension (distance:0 meaning no bytes are skipped between the two content matches), keeping the signature generic enough to catch any HTML file upload rather than being tied to one specific filename. This design choice ensures the rule remains useful even if the malware generates files with different names in future infections. The rule is classified under trojan-activity and references the Malpedia entry for Agent Tesla. The metadata reflects that this is an exfiltration attempt mapped to MITRE ATT&CK tactic TA0010 (Exfiltration) and technique T1048 (Exfiltration Over Alternative Protocol), with a high confidence and major severity rating. Since this rule has not yet been validated in a production environment, its status is set to testing.

Results:

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
12/04/2024-16:20:56.211571  [**] [1:1000002:1] Agent Tesla FTP Upload HTML Exfil File [**] [Classif
ication: A Network Trojan was detected] [Priority: 1] {TCP} 10.12.4.101:49771 -> 192.254.225.136:21
12/04/2024-16:20:56.211571  [**] [1:1000002:1] Agent Tesla FTP Upload HTML Exfil File [**] [Classif
ication: A Network Trojan was detected] [Priority: 1] {TCP} 10.12.4.101:49889 -> 192.254.225.136:21
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2024-12-04T16:20:56.211571-0500",
  "flow_id": 1708112361074766,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49771,
  "dest_ip": "192.254.225.136",
  "dest_port": 21,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "stream (flow timeout)",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000002,
    "rev": 1,
    "signature": "Agent Tesla FTP Upload HTML Exfil File",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "high"
      ],
      "created_at": [
```

```
{
  "timestamp": "2024-12-04T16:20:56.211571-0500",
  "flow_id": 1090967654169465,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49889,
  "dest_ip": "192.254.225.136",
  "dest_port": 21,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "stream (flow timeout)",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000002,
    "rev": 1,
    "signature": "Agent Tesla FTP Upload HTML Exfil File",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "high"
      ],
      "created_at": [
        "2024_12_04"
      ],
```

```
"app_proto": "ftp",
"direction": "to_server",
"flow": {
  "pkts_toserver": 12,
  "pkts_toclient": 17,
  "bytes_toserver": 804,
  "bytes_toclient": 1550,
  "start": "2024-12-04T16:41:07.581690-0500",
  "src_ip": "10.12.4.101",
  "dest_ip": "192.254.225.136",
  "src_port": 49889,
  "dest_port": 21
},
```

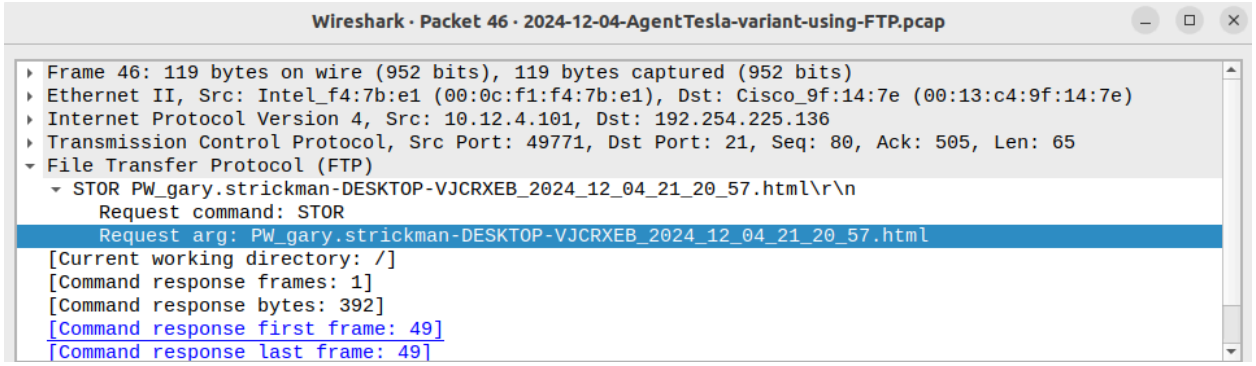

The rule triggered two alerts, both originating from the same internal machine at 10.12.4.101 and destined for the same external FTP server at 192.254.225.136 on port 21, which is the standard FTP control port. The two alerts correspond to two separate TCP connections (identified by different source ports, 49771 and 49889, and different flow IDs), meaning Agent Tesla initiated two distinct FTP sessions to upload HTML files, which is consistent with what was observed in Wireshark.

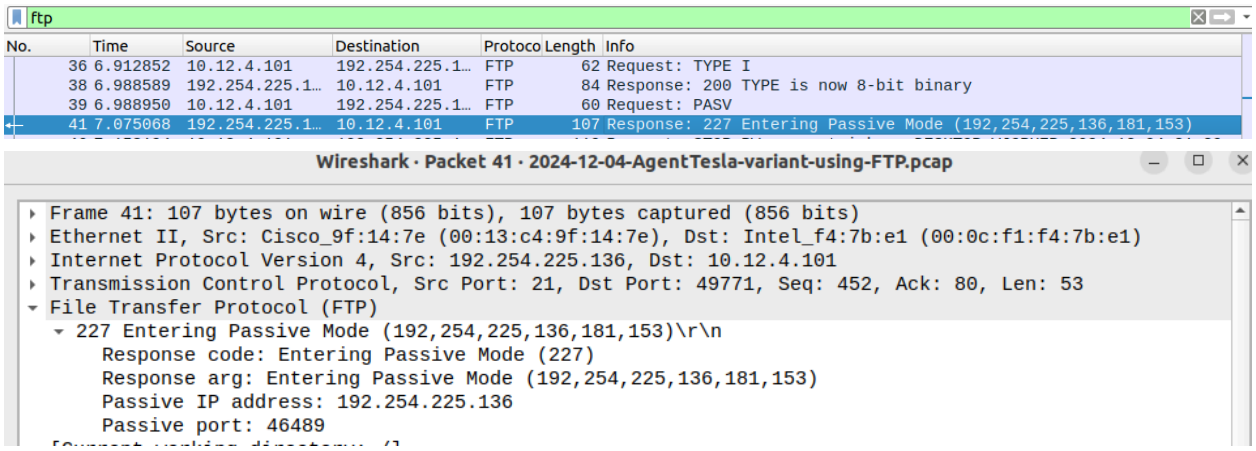
Both alerts fired at the same recorded timestamp (2024-12-04T16:20:56) but the flow data shows the sessions actually started at different times — the first around 16:21 and the second around 16:41 — suggesting the malware performed two upload operations roughly twenty minutes apart. The pkt_src: stream (flow timeout) field indicates Suricata detected the match upon flow reassembly rather than on a single raw packet, which is normal behavior for FTP session analysis.

The flow statistics confirm genuine bidirectional communication took place in both cases, with packets and bytes exchanged in both directions, ruling out false positives caused by one-sided traffic. The metadata fields in the alert correctly reflect the rule's classification, mapping the activity to MITRE ATT&CK tactic TA0010 (Exfiltration) and technique T1048 (Exfiltration Over Alternative Protocol).

5.3: Write a rule to detect the transmission of the file "PW_gary.strickman.." over the FTP-DATA protocol

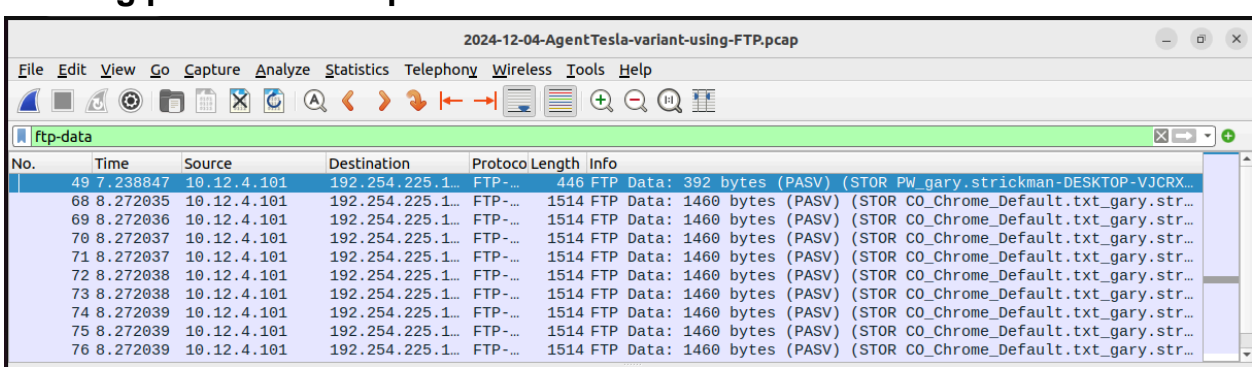
Analysing the FTP packets:



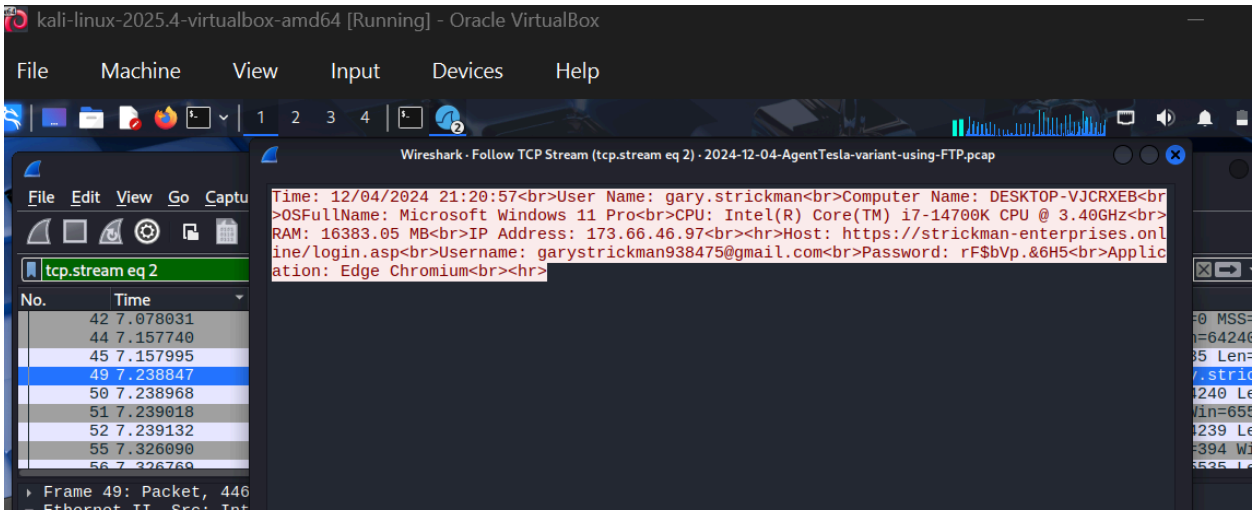


This gives the FTP-DATA IP & port

Filtering packets with ftp-data filter:



Checking the raw file contents:



After the FTP STOR command is issued, the file is not transmitted over the control channel but through a separate FTP-DATA connection established using passive mode (PASV).
The server responds with a 227 Entering Passive Mode message, which specifies the IP address and port used for the data transfer.
Filtering on ftp-data reveals the actual file transmission. By following the TCP stream, we can observe the raw content of the exfiltrated file, confirming that sensitive data is being sent from the infected host to the remote server.

Suricata Rule:

```
alert ftp-data any any -> any any (msg:"Agent Tesla FTP-DATA HTML
credentials transfer"; file.data; content:"Username:";
content:"Password:"; nocase; classtype:trojan-activity;
reference:url,malpedia.caad.fkie.fraunhofer.de/details/win.agent_tesl
a; sid:1000003; rev:1; metadata:created_at 2024_12_04,confidence
high,signature_severity critical,mitre_tactic_id
TA0010,mitre_tactic_name Exfiltration,mitre_technique_id
T1048,mitre_technique_name
Exfiltration_Over_Alternative_Protocol,status testing;)
```

This Suricata rule is designed to detect the exfiltration of sensitive credentials over the FTP-DATA protocol, specifically targeting the behavior of the Agent Tesla malware. The rule applies to any source and destination IP/port combination (any any -> any any) and inspects the content of files being transferred over an established FTP data connection (file.data). It looks for file content containing both "Username:" and "Password:", ignoring case (nocase), which is a typical pattern for credential theft in HTML or text-based data files. The classification is set as trojan-activity, reflecting the malicious nature of the transfer. The rule includes a reference to [Malpedia's Agent Tesla entry]() for context and attribution. Metadata is included to facilitate rule management, indicating the creation date (created_at: 2024_12_04), confidence level (high), signature severity (critical), MITRE ATT&CK tactic and technique identifiers (TA0010: Exfiltration, T1048: Exfiltration Over Alternative Protocol), and current lifecycle status (testing).

The rule is uniquely identified by sid:1000003 with a revision number of rev:1. Overall, this signature is designed to be minimally generic to catch FTP-based credential exfiltration attempts while avoiding excessive false positives.

Results:

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
12/04/2024-16:21:03.538340  [**] [1:1000003:1] Agent Tesla FTP-DATA HTML credentials transfer [**] [Classificatio
n: A Network Trojan was detected] [Priority: 1] {TCP} 10.12.4.101:49772 -> 192.254.225.136:46489
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2024-12-04T16:21:03.538340-0500",
  "flow_id": 2088258534508736,
  "parent_id": 1708109888652814,
  "pcap_cnt": 56,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49772,
  "dest_ip": "192.254.225.136",
  "dest_port": 46489,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000003,
    "rev": 1,
    "signature": "Agent Tesla FTP-DATA HTML credentials transfer",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "high"
      ],
      "created_at": [
        "2024_12_04"
      ],
      "mitre_tactic_id": [
        "TA0010"
      ],
      "mitre_tactic_name": [
        "Exfiltration"
```

```
      "mitre_technique_id": [
        "T1048"
      ],
      "mitre_technique_name": [
        "Exfiltration_Over_Alternative_Protocol"
      ],
      "signature_severity": [
        "critical"
      ],
      "status": [
        "testing"
      ]
    }
  },
  "ftp_data": {
    "filename": "PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html",
    "command": "STOR"
  },
  "files": [
    {
      "filename": "PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html",
      "gaps": false,
      "state": "CLOSED",
      "stored": false,
      "size": 392,
      "tx_id": 0
    }
  ],
  "app_proto": "ftp-data",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 5,
    "pkts_toclient": 4,
    "bytes_toserver": 674,
    "bytes_toclient": 220,
```

```
    "start": "2024-12-04T16:21:03.289602-0500",
    "src_ip": "10.12.4.101",
    "dest_ip": "192.254.225.136",
    "src_port": 49772,
    "dest_port": 46489
  }
}
```


The Suricata rule successfully detected the exfiltration of credentials via FTP-DATA, as seen in both fast.log and eve.json. In fast.log, the alert indicates that a file transfer matching the signature was observed from 10.12.4.101:49772 to 192.254.225.136:46489, with the message "Agent Tesla FTP-DATA HTML credentials transfer", classified as a network Trojan with priority 1. The eve.json output provides a more detailed view, confirming the detection of a file named PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html being sent over FTP-DATA using the STOR command. The alert metadata shows high confidence, critical signature severity, and references the MITRE ATT&CK tactic Exfiltration (TA0010) and technique Exfiltration Over Alternative Protocol (T1048). The flow section in eve.json indicates that 5 packets totaling 674 bytes were sent to the server, matching the expected FTP-DATA activity. Overall, these results confirm that the rule effectively identified Agent Tesla exfiltrating HTML-based credential files via FTP, validating that the file.data inspection and dual content match are functioning correctly.

5.4: Write a rule to detect the upload of the four files over the FTP protocol using PCRE.

Suricata Rule:

```
alert ftp any any -> any any (msg:"Agent Tesla multiple file upload (PCRE)"; content:"STOR"; nocase; pcre:"/STOR\s+(PW_|CO_|KL_)/i"; classtype:trojan-activity; reference:url,malware-traffic-analysis.net; metadata:created_at 2026_03_19, confidence medium, signature_severity medium, mitre_tactic_id TA0010, mitre_tactic_name Exfiltration, mitre_technique_id T1048, mitre_technique_name Exfiltration Over Alternative Protocol, status testing; sid:1000004; rev:2;)
```

This rule uses a PCRE (Perl Compatible Regular Expression) to detect multiple file uploads associated with Agent Tesla. It matches FTP STOR commands followed by filenames beginning with “PW_”, “CO_”, or “KL_”, which correspond to different types of stolen data. This approach increases detection coverage while maintaining flexibility. The confidence is set to medium as the pattern is broader and may match legitimate activity in rare cases.

Results:

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
12/04/2024-16:20:56.211571  [**] [1:1000004:2] Agent Tesla multiple file upload (PCRE) [**] [Classification: A Network Tr
ojan was detected] [Priority: 1] {TCP} 10.12.4.101:49889 -> 192.254.225.136:21
12/04/2024-16:20:56.211571  [**] [1:1000004:2] Agent Tesla multiple file upload (PCRE) [**] [Classification: A Network Tr
ojan was detected] [Priority: 1] {TCP} 10.12.4.101:49771 -> 192.254.225.136:21
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2024-12-04T16:20:56.211571-0500",
  "flow_id": 1090968810433883,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49889,
  "dest_ip": "192.254.225.136",
  "dest_port": 21,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "stream (flow timeout)",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000004,
    "rev": 2,
    "signature": "Agent Tesla multiple file upload (PCRE)",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "medium"
      ],
      "created_at": [
        "2026_03_19"
      ],
      "mitre_tactic_id": [
        "TA0010"
      ],
      "mitre_tactic_name": [
        "Exfiltration"
      ],
      "mitre_technique_id": [
        "T1048"
```

```
      "mitre_technique_name": [
        "Exfiltration Over Alternative Protocol"
      ],
      "signature_severity": [
        "medium"
      ],
      "status": [
        "testing"
      ]
    }
  },
  "app_proto": "ftp",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 12,
    "pkts_toclient": 17,
    "bytes_toserver": 804,
    "bytes_toclient": 1550,
    "start": "2024-12-04T16:41:07.581690-0500",
    "src_ip": "10.12.4.101",
    "dest_ip": "192.254.225.136",
    "src_port": 49889,
    "dest_port": 21
  }
}
{
  "timestamp": "2024-12-04T16:20:56.211571-0500",
  "flow_id": 1708108529877419,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49771,
  "dest_ip": "192.254.225.136",
  "dest_port": 21,
  "proto": "TCP",
  "ip_v": 4,
```

```
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000004,
    "rev": 2,
    "signature": "Agent Tesla multiple file upload (PCRE)",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "medium"
      ],
      "created_at": [
        "2026_03_19"
      ],
      "mitre_tactic_id": [
        "TA0010"
      ],
      "mitre_tactic_name": [
        "Exfiltration"
      ],
      "mitre_technique_id": [
        "T1048"
      ],
      "mitre_technique_name": [
        "Exfiltration Over Alternative Protocol"
      ],
      "signature_severity": [
        "medium"
      ],
      "status": [
        "testing"
      ]
    }
  },
```

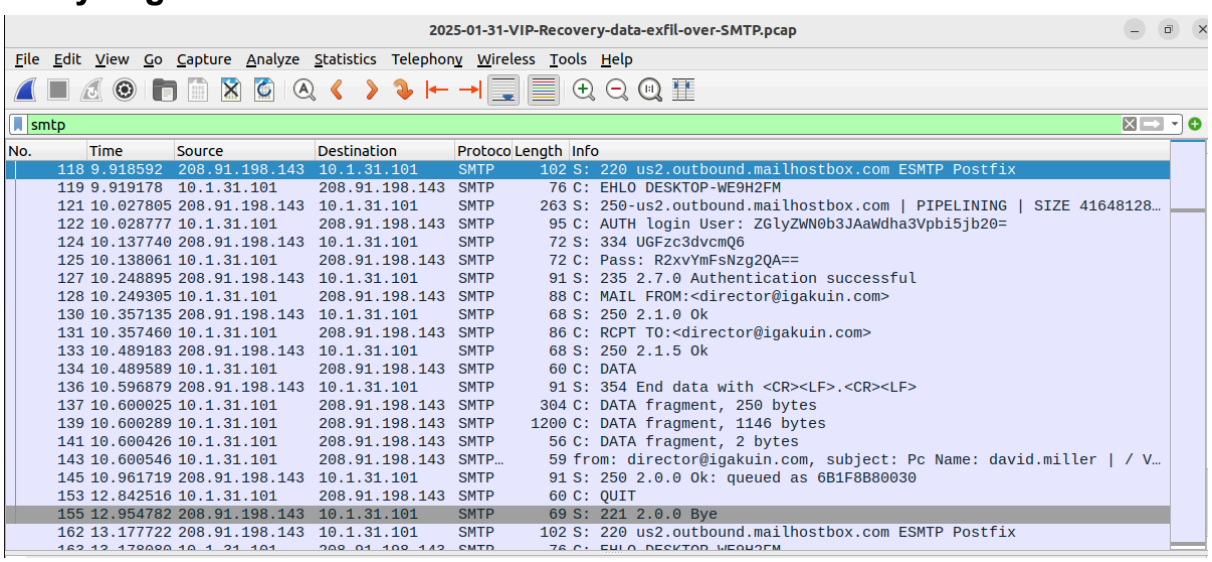
```
  "app_proto": "ftp",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 21,
    "pkts_toclient": 27,
    "bytes_toserver": 1475,
    "bytes_toclient": 2447,
    "start": "2024-12-04T16:21:02.528772-0500",
    "src_ip": "10.12.4.101",
    "dest_ip": "192.254.225.136",
    "src_port": 49771,
    "dest_port": 21
  }
}
```

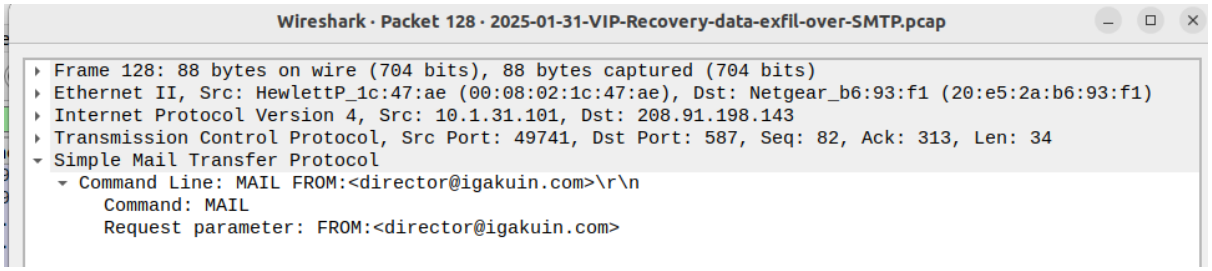

The Suricata rule successfully detected multiple Agent Tesla file uploads over FTP using a PCRE (Perl Compatible Regular Expression) match on the STOR command. In fast.log, two separate alerts were triggered, both flagged as "Agent Tesla multiple file upload (PCRE)" with a classification of A Network Trojan was detected and priority 1. This shows that Suricata

identified multiple FTP sessions from the source IP 10.12.4.101 to the destination 192.254.225.136 on port 21, corresponding to exfiltration attempts.
The eve.json output provides detailed information for each alert. The first alert shows a flow from 10.12.4.101:49889 to 192.254.225.136:21, sending 12 packets (804 bytes) to the server and receiving 17 packets (1550 bytes) from the server. The second alert shows a flow from 10.12.4.101:49771 to the same destination, sending 21 packets (1475 bytes) and receiving 27 packets (2447 bytes). Both alerts confirm traffic moving to_server, matching the STOR upload command in FTP.
Metadata in the alerts confirms a medium confidence level and medium signature severity, and maps the activity to the MITRE ATT&CK tactic Exfiltration (TA0010) and technique Exfiltration Over Alternative Protocol (T1048). These results indicate that the rule correctly matched Agent Tesla's FTP upload patterns, detecting multiple file exfiltration events in the captured traffic.

5.5: Write a rule to detect the SMTP exfiltration performed by Agent Tesla.

Analysing the SMTP Packets:





Wireshark · Packet 175 · 2025-01-31-VIP-Recovery-data-exfil-over-SMTP.pcap

```
▸ Frame 175: 86 bytes on wire (688 bits), 86 bytes captured (688 bits)
▸ Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)
▸ Internet Protocol Version 4, Src: 10.1.31.101, Dst: 208.91.198.143
▸ Transmission Control Protocol, Src Port: 49742, Dst Port: 587, Seq: 116, Ack: 327, Len: 32
▾ Simple Mail Transfer Protocol
  ▾ Command Line: RCPT TO:<director@igakuin.com>\r\n
      Command: RCPT
      Request parameter: TO:<director@igakuin.com>
```

Wireshark · Packet 137 · 2025-01-31-VIP-Recovery-data-exfil-over-SMTP.pcap

```
▸ Transmission Control Protocol, Src Port: 49741, Dst Port: 587, Seq: 154, Ack: 378, Len: 250
▾ Simple Mail Transfer Protocol
  ▾ Line-based text data (8 lines)
      MIME-Version: 1.0\r\n
      From: director@igakuin.com\r\n
      To: director@igakuin.com\r\n
      Date: 31 Jan 2025 20:23:19 +0000\r\n
      Subject:  Pc Name: david.miller | / VIP Recovery \\r\n
      Content-Type: text/plain; charset=us-ascii\r\n
      Content-Transfer-Encoding: quoted-printable\r\n
      \r\n
```

Wireshark · Packet 139 · 2025-01-31-VIP-Recovery-data-exfil-over-SMTP.pcap

```
▸ Transmission Control Protocol, Src Port: 49741, Dst Port: 587, Seq: 404, Ack: 378, Len: 1146
▾ Simple Mail Transfer Protocol
  ▾ Line-based text data (17 lines)
      PW | david.miller | VIP Recovery=0D=0A =0D=0A=0D=0APC Name:DESKTO=\r\n
      P-WE9H2FM=0D=0ADate and Time: 1/31/2025 / 8:23:08 PM=0D=0AClient =\r\n
      IP: 34.205.4.78=0D=0A=0D=0ACountry Name: United States=0D=0ACount=\r\n
      ryCode: US=0D=0ARegion Name: Virginia=0D=0ARegion Code: VA=0D=0AC=\r\n
      ity: Ashburn=0D=0ATimeZone: America/New_York=0D=0ALatitude: 39.04=\r\n
      69=0D=0ALongitude: -77.4903=0D=0AStub Version: 4.4=0D=0A=0D=0A=0D=0A=\r\n
      -------- / VIP Recovery \ --------=0D=0ARecovered From: Google Ch=\r\n
      rome=0D=0AHost: https://amazon.com/=0D=0AUSR: david.miller@miller=\r\n
      shomebrew.com=0D=0APSWD: P@ssw0rd1!=0D=0A-----------------------=\r\n
      ---------=0D=0A =0D=0A-------- / VIP Recovery \ --------=0D=0ARec=\r\n
```

Sensitive information present in plain text in data fragments.

Wireshark · Follow TCP Stream (tcp.stream eq 3) · 2025-01-31-VIP-Recovery-data-exfil-over-SMTP....

```
220 us2.outbound.mailhostbox.com ESMTP Postfix
EHLO DESKTOP-WE9H2FM
250-us2.outbound.mailhostbox.com
250-PIPELINING
250-SIZE 41648128
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN LOGIN
250-AUTH=PLAIN LOGIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250 CHUNKING
AUTH login ZGlyZWN0b3JAaWdha3Vpbi5jb20=
334 UGFzc3dvcmQ6
R2xvYmFsNzg2QA==
235 2.7.0 Authentication successful
MAIL FROM:<director@igakuin.com>
250 2.1.0 Ok
RCPT TO:<director@igakuin.com>
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
MIME-Version: 1.0
From: director@igakuin.com
To: director@igakuin.com
Date: 31 Jan 2025 20:23:19 +0000
Subject:  Pc Name: david.miller | / VIP Recovery \
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: quoted-printable

PW | david.miller | VIP Recovery=0D=0A =0D=0A=0D=0APC Name:DESKTO=
```

```
PW | david.miller | VIP Recovery=0D=0A =0D=0A=0D=0APC Name:DESKTO=
P-WE9H2FM=0D=0ADate and Time: 1/31/2025 / 8:23:08 PM=0D=0AClient =
IP: 34.205.4.78=0D=0A=0D=0ACountry Name: United States=0D=0ACount=
ryCode: US=0D=0ARegion Name: Virginia=0D=0ARegion Code: VA=0D=0AC=
ity: Ashburn=0D=0ATimeZone: America/New_York=0D=0ALatitude: 39.04=
69=0D=0ALongitude: -77.4903=0D=0AStub Version: 4.4=0D=0A=0D=0A=0D=0A=
-------- / VIP Recovery \ --------=0D=0ARecovered From: Google Ch=
rome=0D=0AHost: https://amazon.com/=0D=0AUSR: david.miller@miller=
shomebrew.com=0D=0APSWD: P@ssw0rd1!=0D=0A-----------------------=
---------=0D=0A =0D=0A-------- / VIP Recovery \ --------=0D=0ARec=
overed From: Edge Chromium=0D=0AHost: https://linkedin.com/=0D=0A=
USR: david.miller@millershomebrew.com=0D=0APSWD: P@ssw0rd!=0D=0A-=
---------------------------------=0D=0A =0D=0A-------- / VIP Recov=
ery \ --------=0D=0ARecovered From: Edge Chromium=0D=0AHost: http=
s://facebook.com/=0D=0AUSR: 3841584372=0D=0APSWD: P@ssw0rd!=0D=0A=
----------------------------------=0D=0A =0D=0A=0D=0A=0D=0A=0D=0A=0D=0A=
--------------------------------------------------

.
250 2.0.0 Ok: queued as 6B1F8B80030
QUIT
221 2.0.0 Bye
```

Email headers along with the complete subject and body are visible.

Suricata Rule:

```
alert smtp $HOME_NET any -> $EXTERNAL_NET 587 (msg:"AgentTesla SMTP
Exfiltration-Detection";flow:established,to_server;content:"Subject|3
a|";content:"VIP          Recovery";nocase;pcre:"/\b\w+\s*:\s*\S+/i";
```

```
classtype:trojan-activity;sid:1000008;rev:3;reference:url,malpedia.caad.fkie.fraunhofer.de/details/win.agent_tesla;metadata:created_at 2026_03_19, confidence high, signature_severity critical, status testing, mitre_tactic_id TA0010, mitre_tactic_name Exfiltration, mitre_technique_id T1048, mitre_technique_name Exfiltration_Over_Alternative_Protocol;)
```

This rule detects SMTP exfiltration activity associated with the AgentTesla malware. It specifically monitors outbound SMTP traffic from the internal network ($HOME_NET) to external servers ($EXTERNAL_NET) on port 587. The rule triggers on established SMTP sessions (flow:established,to_server) that contain a subject line including the text "VIP Recovery" (case-insensitive).
Additionally, a regular expression (pcre:"/\b\w+\s*:\s*\S+/i") is used to match any key-value style data in the message body, which allows detection of a wide range of exfiltrated information beyond just usernames and passwords. This makes the rule more flexible in capturing different types of sensitive data being sent by the malware.
The rule is classified as trojan-activity, with a high confidence and critical severity rating. Metadata references MITRE ATT&CK Tactic TA0010 (Exfiltration) and Technique T1048 (Exfiltration Over Alternative Protocol), indicating the type of adversary behavior being detected. The detection focus is on subject containing "VIP Recovery" and any key-value pairs in the body, which may indicate data exfiltration

Results:

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
01/31/2025-15:23:21.959708  [**] [1:1000008:3] AgentTesla SMTP Exfiltration - Broad Detection [**] [Classification: A
Network Trojan was detected] [Priority: 1] {TCP} 10.1.31.101:49741 -> 208.91.198.143:587
01/31/2025-15:23:22.889401  [**] [1:1000008:3] AgentTesla SMTP Exfiltration - Broad Detection [**] [Classification: A
Network Trojan was detected] [Priority: 1] {TCP} 10.1.31.101:49742 -> 208.91.198.143:587
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2025-01-31T15:23:21.959708-0500",
  "flow_id": 1839891397658357,
  "pcap_cnt": 156,
  "event_type": "alert",
  "src_ip": "10.1.31.101",
  "src_port": 49741,
  "dest_ip": "208.91.198.143",
  "dest_port": 587,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000008,
    "rev": 3,
    "signature": "AgentTesla SMTP Exfiltration - Broad Detection",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "high"
      ],
      "created_at": [
        "2026_03_19"
      ],
      "mitre_tactic_id": [
        "TA0010"
      ],
      "mitre_tactic_name": [
        "Exfiltration"
```

```
  },
  "app_proto": "smtp",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 15,
    "pkts_toclient": 21,
    "bytes_toserver": 2384,
    "bytes_toclient": 1567,
    "start": "2025-01-31T15:23:18.559455-0500",
    "src_ip": "10.1.31.101",
    "dest_ip": "208.91.198.143",
    "src_port": 49741,
    "dest_port": 587
  }
}
{
  "timestamp": "2025-01-31T15:23:22.889401-0500",
  "flow_id": 464956859533924,
  "pcap_cnt": 198,
  "event_type": "alert",
  "src_ip": "10.1.31.101",
  "src_port": 49742,
  "dest_ip": "208.91.198.143",
  "dest_port": 587,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000008,
    "rev": 3,
    "signature": "AgentTesla SMTP Exfiltration - Broad Detection",
    "category": "A Network Trojan was detected",
```

```
        "TA0010"
      ],
      "mitre_tactic_name": [
        "Exfiltration"
      ],
      "mitre_technique_id": [
        "T1048"
      ],
      "mitre_technique_name": [
        "Exfiltration_Over_Alternative_Protocol"
      ],
      "signature_severity": [
        "critical"
      ],
      "status": [
        "testing"
      ]
    }
  },
  "app_proto": "smtp",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 20,
    "pkts_toclient": 20,
    "bytes_toserver": 17542,
    "bytes_toclient": 1461,
    "start": "2025-01-31T15:23:21.960224-0500",
    "src_ip": "10.1.31.101",
    "dest_ip": "208.91.198.143",
    "src_port": 49742,
    "dest_port": 587
  }
}
```


The Suricata alerts show that the broad AgentTesla SMTP exfiltration detection rule successfully identified two separate malicious SMTP sessions originating from the internal host 10.1.31.101 to the external server 208.91.198.143 on port 587. The two alerts correspond to two distinct flows, as indicated by different source ports (49741 and 49742) and unique flow IDs.

Each alert is labeled as a network trojan activity (classtype: trojan-activity) with critical severity. The alerts were triggered because the SMTP traffic contained a subject line with "VIP Recovery" and key-value style data in the message body, which matches the broad detection rule. This shows that the malware is attempting to exfiltrate data, potentially including usernames, passwords, or other sensitive information, over an SMTP channel.
The flow details confirm bidirectional traffic with the external server, providing packet and byte counts to support the alert, and the alerts include MITRE ATT&CK references:

- Tactic: TA0010 – Exfiltration
- Technique: T1048 – Exfiltration Over Alternative Protocol

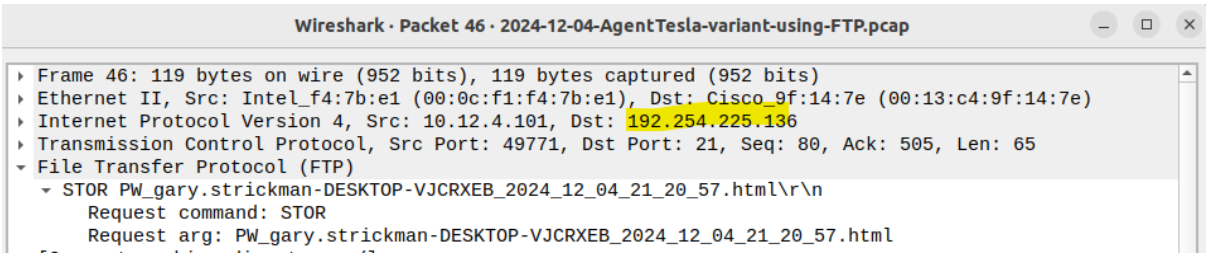
Overall, the detection demonstrates that the enhanced, broad rule effectively captures multiple exfiltration attempts by AgentTesla without being limited to specific key-value pairs, providing better coverage and reducing the risk of missing variants of the malware.

Filestore or tag

6.1: Complete the rule from question 5.2 so that it captures traffic destined for the malicious ip by using a TAG

Identifying the malicious IP:

Opening PCAP in Wireshark and filtering the FTP traffic to look at the STOR commands that upload the file



Destination ip: 192.254.255.136

Suricata Rule:

```
alert ftp any any -> 192.254.225.136 any (msg:"Agent Tesla FTP Upload HTML Exfil File - TAG"; flow:established,to_server; content:"STOR"; nocase; content:".html"; distance:0; tag:session,100,packets; classtype:trojan-activity; reference:url,malpedia.caad.frike.fraunhofer.de/details/win.agent_tesla; sid:1000003; rev:1; metadata:created_at 2024_12_04, confidence high, signature_severity major, mitre_tactic_id TA0010, mitre_tactic_name Exfiltration, mitre_technique_id T1048, mitre_technique_name Exfiltration_Over_Alternative_Protocol, status testing;)
```

This rule is an extension of the rule from 5.2, with two key modifications. First, the destination IP has been narrowed from any to the known malicious FTP server at 192.254.225.136, making the rule more precise and reducing the likelihood of false positives on legitimate FTP traffic elsewhere in the network. Second, the tag:session,100,packets keyword has been added, which instructs Suricata to capture the next 100 packets of the session after the rule fires. This allows the full recovery of FTP session traffic for forensic purposes, providing more context than the alert log alone.

```
bisma@bisma-VirtualBox:~$ cat logs/fast.log
12/04/2024-16:20:56.211571  [**] [1:1000003:1] Agent Tesla FTP Upload HTML Exfil File - TAG [**] [Classification: A
Network Trojan was detected] [Priority: 1] {TCP} 10.12.4.101:49889 -> 192.254.225.136:21
12/04/2024-16:20:56.211571  [**] [1:1000003:1] Agent Tesla FTP Upload HTML Exfil File - TAG [**] [Classification: A
Network Trojan was detected] [Priority: 1] {TCP} 10.12.4.101:49771 -> 192.254.225.136:21
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
```

```
bisma@bisma-VirtualBox:~$ jq 'select(.event_type=="alert")' logs/eve.json
{
  "timestamp": "2024-12-04T16:20:56.211571-0500",
  "flow_id": 1090968101061740,
  "event_type": "alert",
  "src_ip": "10.12.4.101",
  "src_port": 49889,
  "dest_ip": "192.254.225.136",
  "dest_port": 21,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "stream (flow timeout)",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000003,
    "rev": 1,
    "signature": "Agent Tesla FTP Upload HTML Exfil File - TAG",
    "category": "A Network Trojan was detected",
    "severity": 1,
    "metadata": {
      "confidence": [
        "high"
      ],
      "created_at": [
        "2024_12_04"
      ],
      "mitre_tactic_id": [
        "TA0010"
```

```
  },
  "app_proto": "ftp",
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 21,
    "pkts_toclient": 27,
    "bytes_toserver": 1475,
    "bytes_toclient": 2447,
    "start": "2024-12-04T16:21:02.528772-0500",
    "src_ip": "10.12.4.101",
    "dest_ip": "192.254.225.136",
    "src_port": 49771,
    "dest_port": 21
  },
  "capture_file": "logs/log.pcap.1733347256"
}
```


Same results as 5.2, but this time captured packets are stored in the logs subdirectory.

For the TAG keyword to function correctly, configuration changes are required in /etc/suricata/suricata.yaml. The pcap-log output must be enabled so that the tagged packets are actually written to disk. Without these changes, the TAG keyword will fire but no traffic will be stored, rendering the capture feature ineffective. The rest of the rule's metadata, classification and detection logic remain identical to 5.2, as the underlying threat being detected is the same.

```
GNU nano 6.2                                        /etc/suricata/suricata.yaml
# In multi mode the filename takes a few special variables:
# - %n -- thread number
# - %i -- thread id
# - %t -- timestamp (secs or secs.usecs based on 'ts-format'
# E.g. filename: pcap.%n.%t
#
# Note that it's possible to use directories, but the directories are not
# created by Suricata. E.g. filename: pcaps/%n/log.%s will log into the
# per thread directory.
#
# Also note that the limit and max-files settings are enforced per thread.
# So the size limit when using 8 threads with 1000 MiB files and 2000 files
# is: 8*1000*2000 ~ 16TiB.
#
# By default all packets are logged except:
# - TCP streams beyond stream.reassembly.depth
# - encrypted streams after the key exchange
#
- pcap-log:
    enabled: yes
    filename: log.pcap

    # File size limit.  Can be specified in kb, mb, gb.  Just a number
    # is parsed as bytes.
    limit: 1000 MiB

    # If set to a value, ring buffer mode is enabled. Will keep maximum of
    # "max-files" of size "limit"
    max-files: 2000

    # Compression algorithm for pcap files. Possible values: none, lz4.
    # Note also that on Windows, enabling compression will *increase* disk I/O.
    compression: none
```

Pcap file created in the logs directory, tagged sessions stored:

```
bisma@bisma-VirtualBox:~$ ls logs/
eve.json  fast.log  log.pcap.1733347256  pcaplog_stats.log  stats.log  suricata.log
```

Only tagged traffic is captured:

log.pcap.1733347256

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter … <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.12.4.101	10.12.4.1	DNS	73	Standard query 0x3de7 A api.ipify.org
2	0.039953	10.12.4.1	10.12.4.101	DNS	121	Standard query response 0x3de7 A api.ipify.org A
3	0.049197	10.12.4.101	172.67.74.152	TCP	66	49770 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460
4	0.100655	172.67.74.152	10.12.4.101	TCP	58	443 → 49770 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=
5	0.101014	10.12.4.101	172.67.74.152	TCP	54	49770 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.113650	10.12.4.101	172.67.74.152	TLSv1.2	223	Client Hello
7	0.113837	172.67.74.152	10.12.4.101	TCP	54	443 → 49770 [ACK] Seq=1 Ack=170 Win=64240 Len=0
8	0.163280	172.67.74.152	10.12.4.101	TLSv1.2	1514	Server Hello
9	0.163287	172.67.74.152	10.12.4.101	TLSv1.2	1317	Certificate, Server Key Exchange, Server Hello Do
10	0.163620	10.12.4.101	172.67.74.152	TCP	54	49770 → 443 [ACK] Seq=170 Ack=2724 Win=65535 Len=
11	0.167172	10.12.4.101	172.67.74.152	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypte
12	0.167335	172.67.74.152	10.12.4.101	TCP	54	443 → 49770 [ACK] Seq=2724 Ack=263 Win=64240 Len=
13	0.219439	172.67.74.152	10.12.4.101	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted
14	0.240564	10.12.4.101	172.67.74.152	TLSv1.2	238	Application Data
15	0.240732	172.67.74.152	10.12.4.101	TCP	54	443 → 49770 [ACK] Seq=2982 Ack=447 Win=64240 Len=
16	0.344858	172.67.74.152	10.12.4.101	TLSv1.2	526	Application Data
17	0.394018	10.12.4.101	172.67.74.152	TCP	54	49770 → 443 [ACK] Seq=447 Ack=3454 Win=64805 Len=
18	6.122416	10.12.4.101	10.12.4.1	DNS	80	Standard query 0xabe1 A ftp.ercolina-usa.com
19	6.315643	10.12.4.1	10.12.4.101	DNS	110	Standard query response 0xabe1 A ftp.ercolina-usa

6.2: Complete the rule from question 5.3 to retrieve the file "PW_gary.strickman.." using FILESTORE

Suricata Rule:

```
alert ftp-data any any -> any any (msg:"Agent Tesla FTP-DATA HTML
credentials transfer"; file.data; content:"Username:";
content:"Password:"; nocase; filestore; classtype:trojan-activity;
reference:url,malpedia.caad.fkie.fraunhofer.de/details/win.agent_tesl
a; sid:1000003; rev:1; metadata:created_at 2024_12_04,confidence
high,signature_severity critical,mitre_tactic_id
TA0010,mitre_tactic_name Exfiltration,mitre_technique_id
T1048,mitre_technique_name
Exfiltration_Over_Alternative_Protocol,status testing;)
```

This rule is a direct extension of the rule from question 5.3, with the addition of the filestore keyword. The detection logic remains identical, it monitors FTP-DATA traffic for file transfers containing both Username: and Password: strings using the file.data buffer to inspect actual file contents. The key addition is filestore, which instructs Suricata to save the transferred file to disk when the rule fires, allowing the analyst to recover the full exfiltrated file for forensic analysis rather than relying solely on the alert log.

For the **filestore** keyword to function correctly, a configuration change is required in /etc/suricata/suricata.yaml. By default, the file-store section has enabled: no, which must be changed to enabled: yes. Without this change, Suricata will still generate the alert but will not write any files to disk, making the filestore feature ineffective. After enabling it and rerunning Suricata, the exfiltrated file is saved in the logs/filestore/ directory.

```
┌──(kali㉿kali)-[~/logs]
└─$ sed -n '493,500p' /etc/suricata/suricata.yaml
  # To prune the filestore directory see the "suricatactl filestore
  # prune" command which can delete files over a certain age.
  - file-store:
      version: 2
      enabled: yes

      # Set the directory for the filestore. Relative pathnames
      # are contained within the "default-log-dir".

┌──(kali㉿kali)-[~/logs]
└─$
```

```
┌──(kali㉿kali)-[~/logs]
└─$ sudo suricata -c /etc/suricata/suricata.yaml -r ~/pcap/2024-12-04-AgentTesla-variant-using-FTP.pcap -l ~/logs -S ~/TP2.rules
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 3 newly cached: 0 total cacheable: 3
i: threads: Threads created → RX: 1 W: 2 FM: 1 FR: 1   Engine started.
i: suricata: Signal Received.  Stopping engine.
i: pcap: read 1 file, 182 packets, 53464 bytes
```

```
┌──(kali㉿kali)-[~]
└─$ grep 1000003 logs1/fast.log
12/04/2024-16:21:03.538340  [**] [1:1000003:1] Agent Tesla FTP-DATA HTML credentials transfer [**] [Classification: A Network Trojan was detected] [
Priority: 1] {TCP} 10.12.4.101:49772 -> 192.254.225.136:46489

┌──(kali㉿kali)-[~]
└─$ grep 1000003 logs2/fast.log
12/04/2024-16:21:03.538340  [**] [1:1000003:1] Agent Tesla FTP-DATA HTML credentials transfer [**] [Classification: A Network Trojan was detected] [
Priority: 1] {TCP} 10.12.4.101:49772 -> 192.254.225.136:46489
```

The extracted file was then located in the logs/filestore/ directory, organized into subdirectories based on the file hash.

```
┌──(kali㉿kali)-[~/logs]
└─$ sudo ls ~/logs/filestore/
00  07  0e  15  1c  23  2a  31  38  3f  46  4d  54  5b  62  69  70  77  7e  85  8c  93  9a  a1  a8  af  b6  bd  c4  cb  d2  d9  e0  e7  ee  f5  fc
01  08  0f  16  1d  24  2b  32  39  40  47  4e  55  5c  63  6a  71  78  7f  86  8d  94  9b  a2  a9  b0  b7  be  c5  cc  d3  da  e1  e8  ef  f6  fd
02  09  10  17  1e  25  2c  33  3a  41  48  4f  56  5d  64  6b  72  79  80  87  8e  95  9c  a3  aa  b1  b8  bf  c6  cd  d4  db  e2  e9  f0  f7  fe
03  0a  11  18  1f  26  2d  34  3b  42  49  50  57  5e  65  6c  73  7a  81  88  8f  96  9d  a4  ab  b2  b9  c0  c7  ce  d5  dc  e3  ea  f1  f8  ff
04  0b  12  19  20  27  2e  35  3c  43  4a  51  58  5f  66  6d  74  7b  82  89  90  97  9e  a5  ac  b3  ba  c1  c8  cf  d6  dd  e4  eb  f2  f9  tmp
05  0c  13  1a  21  28  2f  36  3d  44  4b  52  59  60  67  6e  75  7c  83  8a  91  98  9f  a6  ad  b4  bb  c2  c9  d0  d7  de  e5  ec  f3  fa
06  0d  14  1b  22  29  30  37  3e  45  4c  53  5a  61  68  6f  76  7d  84  8b  92  99  a0  a7  ae  b5  bc  c3  ca  d1  d8  df  e6  ed  f4  fb

┌──(kali㉿kali)-[~/logs]
└─$
```

```
┌──(kali㉿kali)-[~/logs]
└─$ sudo find ~/logs/filestore/ -type f 2>/dev/null | head -20
/home/kali/logs/filestore/7e/7e07ac68aab2ec158372cb0fe2653fd5b3bf2352540e602a5dd608c8b3a8b07f

┌──(kali㉿kali)-[~/logs]
└─$ sudo cat /home/kali/logs/filestore/7e/7e07ac68aab2ec158372cb0fe2653fd5b3bf2352540e602a5dd608c8b3a8b07f
Time: 12/04/2024 21:20:57<br>User Name: gary.strickman<br>Computer Name: DESKTOP-VJCRXEB<br>OSFullName: Microsoft Windows 11 Pro<br>CPU: Intel(R) Core(
TM) i7-14700K CPU @ 3.40GHz<br>RAM: 16383.05 MB<br>IP Address: 173.66.46.97<br><hr>Host: https://strickman-enterprises.online/login.asp<br>Username: ga
rystrickman938475@gmail.com<br>Password: rF$bVp.86H5<br>Application: Edge Chromium<br><hr>

┌──(kali㉿kali)-[~/logs]
└─$
```

Opening the file confirmed that it contained the full exfiltrated credentials, including:

- **User Name** and **Computer Name** of the victim
- **Operating System** information

- **Public IP Address** of the infected machine
- **Application credentials** including usernames and passwords stolen from browsers

The extracted content matches the FTP data observed in Wireshark, confirming successful exfiltration.

Session Actions Edit View Help


```
"event_type": "fileinfo",
"src_ip": "10.12.4.101",
"src_port": 49772,
"dest_ip": "192.254.225.136",
"dest_port": 46489,
"proto": "TCP",
"ip_v": 4,
"pkt_src": "wire/pcap",
"ftp_data": {
  "filename": "PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html",
  "command": "STOR"
},
"app_proto": "ftp-data",
"fileinfo": {
  "filename": "PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html",
  "sid": [
    1000003
  ],
  "gaps": false,
  "state": "CLOSED",
  "sha256": "7e07ac68aab2ec158372cb0fe2653fd5b3bf2352540e602a5dd608c8b3a8b07f",
  "stored": true,
  "file_id": 1,
  "size": 392,
  "tx_id": 0
}
```


This demonstrates that the filestore keyword successfully recovered the exact file that Agent Tesla exfiltrated, providing analysts with concrete forensic evidence of what data was stolen, which would not have been available from the alert log alone.

Suricata rules management

7.1: Complete the rule from question 5.2 so that it captures traffic destined for the malicious ip by using a TAG

Each rule has been enriched with a standardized set of metadata keywords as specified in the lab instructions. Each rule includes a unique **sid** to identify it unambiguously within the ruleset, paired with a rev number indicating the revision version. The **classtype** keyword categorizes the nature of the detected activity — all rules related to Agent Tesla exfiltration are classified as **trojan-activity**, while the TLS discovery rule (5.1) is classified as **network-scan** to reflect its reconnaissance nature.
A **reference** field is included in each rule pointing to an external source providing context about the threat. For the FTP-based exfiltration rules, this references the Malpedia entry for Agent Tesla, while the PCRE rule references the malware traffic analysis source where the PCAP originates.
The **metadata** field contains several sub-fields. The **created_at** field records the date the rule was written, corresponding to the date of the captured traffic or the date of rule creation. The **confidence** field reflects how reliably the rule detects the intended behavior without false positives — rules targeting specific credential keywords or known domains are rated high, while the broader PCRE rule is rated medium. The **signature_severity** field indicates the criticality of the detected behavior, ranging from low for the IP discovery rule to critical for credential exfiltration rules. The MITRE ATT&CK fields (**mitre_tactic_id, mitre_tactic_name, mitre_technique_id, mitre_technique_name**) map each rule to the corresponding adversary behavior framework, enabling analysts to contextualize alerts within a known attack taxonomy. Finally, all rules carry a **status** of testing, as none of them have been validated in a real production environment. As the rules mature through operational use, this status would progress through stages such as pre-exploitation, exploitation, and eventually disabled if the threat becomes obsolete.
The rules from sections 6.1 and 6.2 are extensions of rules 5.2 and 5.3 respectively, and therefore inherit the same metadata classification. Rule 6.1 adds the tag:session,100,packets keyword to capture full session traffic upon detection, while rule 6.2 adds the filestore keyword to extract and store the exfiltrated file on disk. Since these rules extend existing detection logic rather than introducing new threat signatures, their metadata fields remain consistent with their parent rules — both are classified as trojan-activity with high confidence, mapped to MITRE ATT&CK tactic TA0010 and technique T1048, and carry a status of testing.

Below is a summary table of the metadata applied to each rule:
Rule 5.1: **TLS detection to api.ipify.org** SID: 1000001 | Rev: 2 | Classtype: network-scan | Confidence: high | Severity: low | Tactic: TA0007 (Discovery) | Technique: T1016 (System Network Configuration Discovery) | Status: testing

Rule 5.2: **FTP STOR HTML upload detection** SID: 1000002 | Rev: 1 | Classtype: trojan-activity | Confidence: high | Severity: major | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing

Rule 5.3: **FTP-DATA credential transfer detection** SID: 1000003 | Rev: 1 | Classtype: trojan-activity | Confidence: high | Severity: critical | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing

Rule 5.4: **PCRE multi-file upload detection** SID: 1000004 | Rev: 2 | Classtype: trojan-activity | Confidence: medium | Severity: medium | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing

Rule 5.5: **SMTP exfiltration detection** SID: 1000008 | Rev: 3 | Classtype: trojan-activity | Confidence: high | Severity: critical | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing

Rule 6.1: **FTP TAG session capture** SID: 1000005 | Rev: 1 | Classtype: trojan-activity | Confidence: high | Severity: major | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing

Rule 6.2: **FTP-DATA FILESTORE** file extraction SID: 1000006 | Rev: 1 | Classtype: trojan-activity | Confidence: high | Severity: critical | Tactic: TA0010 (Exfiltration) | Technique: T1048 (Exfiltration Over Alternative Protocol) | Status: testing